

tp3 Experimento 1

DUT : ① (device under test).

$$V_i = 2,76 V_{pp} \text{ (antes de saturar)}$$

$$V_i = 0,89 V_{CF} \quad ; \quad F = 1 \text{ KHz}$$

$$d) \quad V_{o(V_{i0})} = 10,6 V_{pp}$$

$$V_{o(V_{i0})} = 3,75 V_{CF}$$

$$e \text{ y } f) \quad V_o = V_{Rc} = 5,3 V_{pp} \text{ (50\% } V_{o(V_{i0})})$$

$$R_c = 42 \Omega = Z_o$$

Experimento 2:

$$a) \quad V_{o(V_{i0})} = 10,6 V_{pp}$$

$$b \text{ y } c) \quad V_{o(\text{con } R_{in})} = 5,3 V_{pp}$$

$$R_{in} = 1096 \Omega$$

exp 3

$$R_c = 42 \Omega = R_x$$

$$(V_o = V_{Rc}) \Rightarrow 7,2 \text{ dBu}$$

Fotos

Newf.le	Foto
0	V_i V_o } V_{i0}
1	$V_{o(V_{i0})}$

exp. 4

$(v_i) - 1 \text{ dBu}$

exp 5.

(1) $v_i = 40 \text{ mV}$
 $v_o(v_{200}) = 7,2 \text{ Vpp}$

$$A_{oL} = \frac{7,2 \text{ V}}{40 \text{ mV}} = 180$$

(2) $v_s(1 \text{ k}\Omega) = 4,64 \text{ Vpp}$

Fotos Pen Gestan	
NewFile	Foto
0	$v_o(v_{200})$
1	$v_o(1 \text{ k}\Omega)$
2	$v_{oF}(v_{200})$
3	$v_{sF}(1 \text{ k}\Omega)$
4	$v_{sF}(100 \Omega)$
5	$v_{sF}^{(\text{ultimo})}(100 \Omega)$
6	$v_{oF}(v_{200}) (\text{nvero})$
7	$v_{sF}(33 \Omega)$

$$R_o = R_L \left(\frac{v_o}{v_{sF}} - 1 \right) = 1000 \left(\frac{7,2 \text{ Vpp}}{4,64 \text{ Vpp}} - 1 \right) = \boxed{551,7 \Omega}$$

(3) $v_{oF}(v_{200}) = 432 \text{ mV}$ $v_{oF}(1 \text{ k}\Omega) = 416 \text{ mV}$

$$A_{oF} = G = \frac{v_{oF}}{v_i} = \frac{432 \text{ mV}}{40 \text{ mV}} = 10,8$$

$$R_{oF} = R_o' = \frac{R_o \cdot G}{A} = \frac{551,7 \cdot 10,8}{180} = \boxed{33 \Omega}$$

$$\textcircled{4} \quad V_{OF}(33\Omega) = 190\text{ mV}$$

$$R_0' = R_L \left(\frac{r_{o1}}{r_{s1}} - 1 \right) = 33 \left(\frac{432}{190} - 1 \right) = \boxed{42\Omega}$$

exp. 6

$$\textcircled{5} \text{ a) } V_{O_max}(560\Omega) = 3,56\text{ V}$$

$$V_i = 46,4\text{ mV}$$

$$\text{d) } V_{OF_max}(33\Omega) = 464\text{ mV}$$

$$V_i = 100\text{ mV}$$

$$P_{mix} = \frac{V_{pp}^2}{8 \cdot R} = \frac{3,56^2}{8 \cdot 560} = \boxed{2,8\text{ mW}}$$

$$P_{mix(realm)} = \frac{464\text{ mV}^2}{8 \cdot 33\Omega} = \boxed{0,815\text{ mW}}$$

NewFile	Foto
8	$V_{O_max}(560\Omega)$
9	$V_{OF_max}(33\Omega)$
10	$V_o(500\text{ Hz})$
11	$V_o(54\text{ kHz})$
12	$V_o(100\text{ kHz})$
13	$V_o(91\text{ KHz})$
14	$V_o(72\text{ KHz})$
15	$V_o(45\text{ KHz})$
16	$V_o(60\text{ KHz})$
17	$V_o(77\text{ Hz})$

exp. 7

$$V_{O_max}(1\text{ kHz}) = 7,2\text{ Vpp}$$

$$V_o(-3\text{ dB}) = 5,091\text{ Vpp (según chart)}$$

$$V_o(-2\text{ dB}) = 5,72\text{ Vpp}$$

RT2 Frecc. L220 zbiorko

F	V _o
1000hz	7,2Vpp
500hz	6,96Vpp
100hz	6,08Vpp
77hz	5,72Vpp (-2db)

$$F_{cin} = 54hz \quad 5,12Vpp$$

F	V _o
1Khz	7,2Vpp
45Khz	6,64Vpp
60Khz	6,16Vpp
72Khz	5,72Vpp (-2db)

$$F_{csup} = 91,1Khz \quad 5,12Vpp$$

L220 cestwo

$$V_{oF-max}(1Khz) = 8,16Vpp$$

$$V_i = 716mV$$

$$V_{oF}(-2db) = 6,48Vpp$$

$$V_{oF}(-3db) = 5,77Vpp$$

F	V _{oF}
500hz	8,08Vpp
100hz	7,92Vpp
13hz	6,48Vpp (-2db)

$$F_{cinF} = 10hz \quad 5,76Vpp (-3db)$$

Wavefile	Foto
18	V _{oF} (1Khz)
19	V _{oF} (500hz)
20	V _{oF} (100hz)
21	V _{oF} (13hz)
22	V _{oF} (10hz)
23	V _{oF} (1,6MHz)

δ

$f_{c\text{sup}} = 1,587 \text{ MHz}$	$5,76 \text{ Vpp} (-3 \text{ dB})$
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